

GCE AS MARKING SCHEME

PRACTICE PAPER 2 (ADVANCED)

AS ECONOMICS - COMPONENT 2

B520U20-1

GENERAL MARKING GUIDANCE

Positive Marking

Credit should be given for what the learner writes, rather than penalising omissions. It should be possible for a very good response to achieve full marks and a very poor one to achieve zero marks. Marks should not be deducted for a less than perfect answer if it satisfies the criteria of the mark scheme.

Each question has indicative content suggesting the range of economic concepts, theory, issues and arguments which might be included. This is not exhaustive and learners do not have to include all the indicative content to reach the highest level. Level-based mark schemes sub-divide the total mark across individual assessment objectives.

EDUQAS GCE AS ECONOMICS - COMPONENT 2

PRACTICE PAPER 2 (ADVANCED) MARK SCHEME

Q.1 (a) Using a diagram, outline the likely impact of expanded offshore wind investment on the wages of skilled offshore engineers. [4]

Band	AO1 (2 marks) Is the diagram accurate and used?	AO1 (2 marks) Is understanding of the reason clear?
2	2 marks — Good understanding Strong use of labour market diagram showing rightward shift in demand for engineers, leading to higher wages. No meaningful errors.	2 marks — Good understanding Clear and detailed understanding of why demand for skilled offshore engineers changes.
1	1 mark — Limited Weaker diagram with labelling errors or limited scope.	1 mark — Limited Of why demand or supply of labour changes.
0	0 marks Diagram is not valid.	0 marks No valid understanding.

Indicative content:

AO1: The expansion of offshore wind investment increases the derived demand for skilled offshore engineers (engineers are needed to design, install and maintain wind turbines). The demand for labour shifts right from D1 to D2. Since the supply of skilled offshore engineers is relatively inelastic in the short run (training takes years; geographic immobility), wages rise significantly from W1 to W2, with a small increase in employment from Q1 to Q2.

Alternative: a leftward shift in labour supply (due to ageing workforce or migration restrictions) leading to higher wages is also creditable.

Q.1 (b) Discuss the likely cross elasticity of demand between electric vehicles and petrol in the UK.

[6]

Band	AO1 (2 marks) Is XED understood?	AO3 (2 marks) Is the reasoning explained?	AO4 (2 marks) Is the answer debated and judged?
2	2 marks — Good Of XED in terms of substitutes (positive) or complements (negative).	2 marks — Good Developed analysis explaining why EVs and petrol are substitutes (positive XED).	2 marks — Good Reasoned judgement on the likely XED and strength. Counter argument present and developed.
1	1 mark — Limited Partial understanding (e.g. formula only).	1 mark — Limited Underdeveloped chain of reasoning.	1 mark — Limited Counterargument present but no overall judgement.
0	0 marks	0 marks	0 marks

Indicative content:

AO1: Cross elasticity of demand (XED) measures the percentage change in quantity demanded for one good after a change in the price of another. Substitutes have a positive XED, complements have a negative XED.

AO3: EVs and petrol are substitutes — both provide transport. As the price of petrol rises, the running cost of petrol cars increases, making EVs comparatively more attractive. Demand for EVs would rise — a positive XED. However, the XED is likely to be relatively weak in the short run because: (1) EVs are expensive upfront and not easily affordable on short notice; (2) drivers are often locked into existing petrol cars for several years; (3) limited public charging infrastructure deters switching. So XED would be positive but inelastic.

AO4: Strength of XED depends on the size of the petrol price change — a large rise (e.g. £2/litre) might trigger faster switching. Depends on time period — XED is more elastic in the long run as more drivers can replace cars and infrastructure improves. Depends on EV prices — if government subsidies and battery cost reductions make EVs cheaper, XED would strengthen. Depends on income — high-income households can switch more easily. Depends on substitutes such as hybrids and public transport. Overall, XED is likely to be weak positive in the short run but more elastic in the long run, especially as the 2030 ban on new petrol/diesel cars approaches.

Q.1 (c) Using Table 1, explain how the UK's green investment programme can lead to the benefits of the multiplier effect. [6]

Band	AO1 (2 marks) Multiplier effect understood?	AO2 (2 marks) Answer in context?	AO3 (2 marks) How benefits are gained?
2	2 — Good Multiplier as a bigger final increase in GDP than initial injection.	2 — Good application Effective use of data (£300bn investment, £420bn GVA, 480,000 jobs, 60% UK contracts).	2 — Good analysis Developed analysis of how the benefits flow through.
1	1 — Limited	1 — Limited Limited reference to data.	1 — Limited Chain of reasoning underdeveloped.
0	0	0	0

Indicative content:

AO1: The multiplier effect occurs when an initial injection causes a bigger final increase in national income/real GDP.

AO2: £300bn green investment (public and private) is the initial injection. £420bn estimated GVA over 25 years (1.4x multiplier). 480,000 direct green jobs. 60% supply chain contracts to UK firms. 21 million households served. £300 annual saving on bills (long run).

AO3: The £300bn investment in wind, solar, EV infrastructure and insulation drives output in green industries. The 480,000 workers in these industries receive wages, which they spend on goods and services in their communities (consumption) — generating second-round income for retailers, restaurants and service firms. As 60% of contracts go to UK firms, leakages to imports are reduced, keeping more spending domestically. Profitable green firms reinvest, generating further rounds. The £300 annual saving per household on bills boosts disposable income for 21 million households, further raising consumption. The combined demand-side multiplier produces the £420bn estimated GVA, significantly larger than the initial £300bn injection.

Q.1 (d) Evaluate whether a comprehensive carbon tax is better than command-and-control regulation to reduce the negative externalities of carbon emissions in the UK. [10]

Band	AO1 (3 marks) Understanding	AO3 (3 marks) Analysis	AO4 (4 marks) Debate and judgement
3	3 — Excellent Of carbon tax AND regulation.	3 — Excellent Well-developed analysis of both.	4 — Excellent Reasoned judgement with developed counter arguments.
2	2 — Good Of carbon tax OR regulation.	2 — Good Developed analysis of either or both.	2-3 — Good Counter arguments developed.
1	1 — Limited	1 — Limited Underdeveloped or unfocused.	1 — Limited Counterarguments superficial.
0	0	0	0

Indicative content:

AO1: A carbon tax is a market-based instrument that places a per-tonne price on CO₂ emissions, internalising the external cost. Command-and-control regulation uses legal limits or bans (e.g. emission standards, the 2030 petrol/diesel car ban). Negative externalities occur when production or consumption imposes costs on third parties not reflected in market prices, causing welfare loss.

AO3 — Carbon tax: internalises the external cost. Firms with high abatement costs pay the tax; firms with cheap abatement reduce emissions — efficient outcome. Raises revenue (£18bn at £75/tonne) for green spending or compensation. Provides ongoing dynamic incentive to innovate cleaner technology. Moves the market closer to the social optimum (MSC = MSB).

Regulation: provides certainty about emission outcomes (e.g. zero new petrol cars from 2030). Doesn't require complex price-setting. Easier to enforce in some industries. Can target specific high-emission activities directly.

AO4 — Carbon tax weaknesses: Setting the 'right' price is difficult — too low and emissions don't fall enough; too high and competitiveness suffers. Regressive — low-income households spend more of income on energy. Carbon leakage: firms relocate to countries without carbon taxes (mitigated by CBAM). Lobbying can erode coverage (only 28% of UK emissions covered by UK ETS).

Regulation weaknesses: Inefficient — same standard for all firms regardless of abatement costs (deadweight loss). No revenue for government. Limited dynamic incentive once standard is met. Difficult to update as technology evolves. Can be captured by industry interests.

Judgement: Carbon tax is more efficient in theory because it harnesses the price mechanism — firms self-select cheapest abatement. However, regulation is more predictable and politically acceptable for high-profile emissions sources (cars, power stations). A combined approach is most effective: carbon tax for broad coverage with regulation for specific high-emission sectors. Comprehensive carbon tax is likely 'better' for economic efficiency but only if combined with redistribution to address regressivity.

Q.1 (e)(i) Define 'government failure' and explain why renewable energy subsidies could be considered an example of government failure. [4]

Band	AO1 (2 marks) Government failure understood?	AO3 (2 marks) Explanation clear?
2	2 — Good Government intervention to correct market failure creating net welfare loss.	2 — Good Developed analysis of how renewable subsidies could create net welfare loss.
1	1 — Limited	1 — Limited Underdeveloped chain of reasoning.
0	0	0

Indicative content:

AO1: Government failure occurs when government intervention designed to correct market failure creates a net welfare loss greater than the original failure.

AO3: Renewable energy subsidies are intended to correct the negative externality of fossil fuel emissions and the under-provision of public goods like a stable climate. However, they could constitute government failure in several ways: (1) Excessive cost — early UK solar subsidies under the Feed-in Tariff scheme proved more generous than needed, costing consumers billions on energy bills. (2) Information failure — government may pick the wrong technologies (e.g. early biofuel subsidies later found to increase emissions). (3) Unintended consequences — large subsidies for offshore wind have driven up grid costs and electricity bills, particularly harming low-income households (regressive). (4) Regulatory capture — subsidies become entrenched and difficult to phase out, distorting investment. The net welfare loss can exceed the original externality cost. Specifically, the Contracts for Difference scheme has been criticised for paying above-market prices, transferring wealth from consumers to producers without proportionate emissions reductions.

Q.1 (e)(ii) Using the data, assess the extent to which the UK's net zero investment programme would be positive for the short run and long run macroeconomic performance of the UK. [10]

Band	AO2 (3 marks) In context?	AO3 (3 marks) Consequences explained?	AO4 (4 marks) Debated and judged?
3	3 — Excellent Thorough use of data.	3 — Excellent Both short and long run, demand and supply side.	4 — Excellent Reasoned judgement, counter arguments developed.
2	2 — Good	2 — Good Short and/or long run.	2-3 — Good
1	1 — Limited Short run (AD) only.	1 — Limited Demand side only.	1 — Limited
0	0	0	0

Indicative content (answer is reversible):

AO2: £300bn investment; £420bn GVA over 25 years; 480,000 green jobs; 60% UK contracts; 68% CO₂ reduction; £300 annual household saving; £344bn long-run net zero cost.

AO3 — Real GDP: Short run: increase in AD via increased G and I plus the multiplier (£300bn → £420bn). Long run: increase in LRAS through new capital stock (wind farms, EVs, grid upgrades), higher productivity, and reduced exposure to imported fossil fuel price shocks (energy security).

Unemployment: 480,000 jobs in green sectors. Shift from carbon-intensive industries (e.g. North Sea oil) creates short-run structural unemployment but long-run net job creation. Geographic regions (Scotland, North East) benefit from offshore wind investment, addressing regional inequalities.

Inflation: Short run: large G and I push AD right, demand-pull inflation possible. Long run: lower household bills (£300 saving) reduce CPI; greater energy security reduces inflation shocks from global oil/gas markets.

Balance of Payments: 60% UK contracts retain spending domestically, but 40% imports (e.g. solar panels from China) worsen trade balance in short run. Long run, reduced fossil fuel imports improve current account;

emerging green tech exports could become a UK export industry.

AO4: Real GDP gains depend on whether productivity improvements materialise — if green tech is less efficient than fossil fuels (e.g. intermittent renewables needing back-up generation), short-run productivity may fall. Crowding out: £300bn public investment could displace private investment in other sectors. The 480,000 jobs are net only after losses elsewhere — Aberdeen oil workers may lose jobs faster than they retrain.

Inflation: green inflation ('greenflation') is a real risk — high demand for critical minerals (lithium, copper) drives global prices up. The cost of net zero (£344bn) may require higher taxes or borrowing, worsening fiscal position.

Balance of Payments: depends on the UK's competitiveness in emerging clean tech. Currently the UK lags China in solar and battery manufacturing.

Judgement: Overall the programme is likely net positive in the long run, especially as fossil fuel costs rise globally, but the short-run transition will be costly and uneven. Effective targeting of investment, retraining and price support for vulnerable households is essential to realise the gains.

Q.2 (a)(i) Using a PPF diagram, illustrate the impact of an increase in green technological progress on an economy that produces consumer goods and capital goods. [2]

Mark	AO1
2 marks	Accurate drawing showing the original concave PPF and an outward shift of the entire PPF, indicating an increase in productive capacity for both consumer and capital goods.
1 mark	A drawing of a PPF with no outward shift shown.
0 marks	For drawing just axes, or PPF drawn incorrectly.

Indicative content: Concave PPF with consumer goods on one axis and capital goods on the other, showing the original PPF1 and a new PPF2 shifted outward, indicating that technological progress enables higher output of both goods.

Q.2 (a)(ii) Using an AD/AS diagram, explain how expansionary monetary policy can be used to stimulate economic recovery from a recession. [6]

Band	AO1 (2 marks) Diagram correct?	AO1 (2 marks) Monetary policy understood?	AO3 (2 marks) How it stimulates growth?
2	2 — Good Accurate integrated AD/AS diagram showing rightward shift in AD due to expansionary monetary policy.	2 — Good Full understanding of at least two transmission mechanisms.	2 — Good Developed analysis.
1	1 — Limited Diagram has errors or not integrated.	1 — Limited Partial understanding (e.g. only cost of borrowing).	1 — Limited
0	0	0	0

Indicative content:

AO1: Diagram shows AD shifting right ($AD1 \rightarrow AD2$) with price level rising ($P1 \rightarrow P2$) and real GDP rising ($Y1 \rightarrow Y2$). Expansionary monetary policy works through cutting Bank Rate and/or QE.

AO3 — Transmission mechanisms:

Consumption: Lower interest rates reduce the cost of borrowing on credit cards and loans, encouraging credit-financed spending. Lower mortgage costs increase discretionary income. Lower returns on saving encourage consumption over saving.

Investment: Lower business loan costs make more investment projects financially viable (the IRR exceeds the cost of capital). QE injects new central bank reserves, encouraging banks to lend.

Asset prices: Lower rates raise house and share prices, creating positive wealth effects that boost consumption (Pigou effect).

Exchange rate: Lower rates reduce demand for sterling (hot money outflows), depreciating the pound. Cheaper exports and dearer imports improve net exports (X-M).

Confidence: Lower rates and QE signal central bank support, raising consumer and business confidence and animal spirits.

Combined: $AD1 \rightarrow AD2$, $Y1 \rightarrow Y2$ (recovery from recession). The recovery may be accompanied by mild inflation as $P1 \rightarrow P2$.

Q.2 (b)(i) Using the data, describe how the proposed comprehensive carbon tax and green sovereign wealth fund could decrease negative externalities. [4]

Band	AO1 (2 marks) Decrease in negative externalities understood?	AO2 (2 marks) Data used?
2	2 — Good Understanding of negative externality and decrease.	2 — Good Use of BOTH carbon tax AND green sovereign wealth fund.
1	1 — Limited	1 — Limited ONE measure used.
0	0	0

Indicative content:

AO1: A negative externality occurs when production or consumption causes a harmful spillover to third parties. Decreasing emissions reduces climate damage (flooding, heatwaves, biodiversity loss) to UK households and globally.

AO2: A comprehensive carbon tax of £75/tonne raises the private cost of polluting activities, internalising the external cost. Firms reduce emissions to avoid the tax — moving output closer to the socially optimal level. The £18bn annual revenue can be reinvested in mitigation. The £100bn green sovereign wealth fund directly funds renewable energy infrastructure and emerging clean technologies, displacing fossil fuel generation and reducing the external costs of CO₂ emissions. Combined, the tax reduces demand for polluting outputs while the fund increases supply of clean alternatives — a more comprehensive reduction in the negative externality.

Q.2 (b)(ii) Using the data, describe how the proposed comprehensive carbon tax and green sovereign wealth fund could increase positive externalities. [4]

Band	AO1 (2 marks) Positive externalities understood?	AO2 (2 marks) Data used?
2	2 — Good Of positive externality and increase.	2 — Good Use of BOTH measures.
1	1 — Limited	1 — Limited
0	0	0

Indicative content:

AO1: A positive externality occurs when production or consumption causes a benefit to third parties, so social benefit exceeds private benefit.

AO2: The carbon tax revenue (£18bn/year) can fund clean R&D, schools and training — generating positive externalities through a more skilled, healthier workforce. The price signal accelerates innovation in clean technologies which spill over to other firms and countries.

The £100bn green sovereign wealth fund invests in emerging tech (e.g. hydrogen, battery storage), creating R&D; spillovers — when one firm innovates, knowledge diffuses through the wider economy. Better air quality from less fossil fuel combustion improves public health (lower NHS costs, fewer days lost to illness). Green jobs created (480,000+) bring positive externalities through community regeneration in former industrial regions.

Q.2 (c) Using a demand and supply diagram, consider who benefits the most from a decrease in ad valorem VAT on green technologies, the consumer or the firms? [7]

Band	AO1 (3 marks) Diagram	AO3 (2 marks) Impact analysed?	AO4 (2 marks) Debated and judged?
3	3 — Excellent D&S; with pivot shift downward and incidence shown for both consumer and producer (or surplus areas).		
2	2 — Good D&S; with shift but not pivot, or pivot with one incidence.	2 — Good Developed analysis of consumer and producer impact.	2 — Good Reasoned judgement with counter arguments.
1	1 — Limited Parallel shift only.	1 — Limited One side or underdeveloped.	1 — Limited Counter arguments superficial.
0	0	0	0

Indicative content:

AO1: Diagram shows pivot shift of supply curve downward ($S_1 \rightarrow S_2$) since VAT is ad valorem (proportional). Price falls from P_1 to P_2 , quantity rises Q_1 to Q_2 . Consumer benefit = $P_1 - P_2$. Firm benefit = $P_2 - (P_1 - \text{full tax cut})$. Alternatively show consumer surplus and producer surplus areas.

AO3: When demand is inelastic (e.g. for installed solar panels in a market with limited substitutes), firms pass less of the cut on (small fall in price) and retain a larger share as producer surplus. When demand is elastic (e.g. for portable EV chargers with many substitute brands), firms must pass on most of the cut to increase quantity sold, so consumers gain more.

AO4: Depends on PED — green tech is often new with relatively few substitutes (initially), so demand may be inelastic and firms gain more. As markets mature, competition intensifies and PED rises — consumers gain more. Depends on whether firms pass on the cut: if green tech firms have market power (oligopoly in solar panel supply), they may keep more of the VAT cut as profit. PES matters too — if green tech supply is constrained (skills shortages, supply chain bottlenecks), prices fall less and consumers gain less. Depends on whether the government uses revenue from carbon tax to fund the VAT cut — broader welfare effects matter. Overall, in the short run firms in concentrated green tech markets likely gain more; in mature competitive markets consumers gain more.

Q.2 (d) Using the data from Table 1, discuss whether the proposed green tax and revenue measures are a good way to raise government revenue. [9]

Band	AO2 (3 marks)	AO3 (2 marks)	AO4 (4 marks)
3	3 — Excellent Table 1 used fully.		4 — Excellent Reasoned judgement in context.
2	2 — Good	2 — Good Developed analysis.	2-3 — Good
1	1 — Limited	1 — Limited	1 — Limited
0	0	0	0

Indicative content:

AO2: Carbon tax £18bn p.a.; CBAM £2bn; Frequent flyer levy progressive structure; Wealth tax £10bn p.a.; Fossil fuel subsidy reform £8bn. Total potential ~£38bn annually.

AO3: Carbon tax at £75/tonne creates a strong price signal — both raises revenue and reduces emissions (double dividend). CBAM protects UK firms from competitive disadvantage and prevents carbon leakage, supporting domestic green industries. The frequent flyer levy is progressive — targets high-carbon, high-income behaviour. The wealth tax on the top 1% raises £10bn from a small group with low marginal propensity to consume (so minimal AD impact). Fossil fuel subsidy reform removes a £8bn distortion that contradicts climate goals.

AO4: Carbon tax revenues will FALL over time as emissions decline (revenue tension — incentive vs revenue raising). Carbon leakage risk if carbon tax is stronger than international peers — though CBAM helps. CBAM faces WTO challenges and administrative complexity. The frequent flyer levy faces objections (privacy of tracking flights, displacement to international airports). Wealth tax: international evidence (France, Sweden) shows many wealth taxes have failed — high-wealth individuals can relocate or restructure to avoid (£10bn estimate may be very optimistic). Laffer curve concerns. Fossil fuel subsidy reform may raise consumer fuel/gas bills, regressive impact. The £344bn long-run cost of net zero may exceed even these revenues.

Judgement: The carbon tax and CBAM are strong proposals — they raise revenue AND reduce externalities. Fossil fuel subsidy reform is straightforward and long-overdue. The wealth tax is controversial and may underperform projections; the frequent flyer levy is creative but small in revenue. Overall a reasonable revenue-raising package but the wealth tax should be scrutinised carefully. The mix is preferable to a single very high tax which would create greater distortions.

Q.2 (e) Using the data, discuss whether worker retraining programmes or energy bill rebates is the best way to ensure a 'just transition' that reduces inequality during the move to net zero. [8]

Band	AO2 (2 marks)	AO3 (3 marks)	AO4 (3 marks)
3		3 — Excellent Detailed analysis of both.	3 — Excellent Reasoned judgement, counter arguments developed.
2	2 — Good	2 — Good Developed analysis of one OR both.	2 — Good
1	1 — Limited	1 — Limited	1 — Limited
0	0	0	0

Indicative content:

AO2: The carbon transition will displace workers in fossil fuel industries (North Sea oil, gas heating engineers, petrol vehicle manufacturers). 480,000 green jobs are created but not always in the same locations or for the same skill sets. Low-income households spend a higher share of income on energy and are most exposed to rising green energy bills in the short run.

AO3 — Retraining: Closes the structural unemployment gap by enabling former fossil fuel workers to move into renewable energy roles. Builds long-run human capital and labour productivity. Supports regional economies (Aberdeen oil workers → offshore wind technicians). Reduces inequality permanently by upskilling workers rather than just compensating them.

Energy bill rebates: Targeted income support for low-income households directly offsets the regressive impact of carbon tax/higher green energy prices. Quick to administer, no take-up gap if automatic. Preserves consumer choice in how to use the rebate.

AO4 — Retraining limitations: Workers nearing retirement may not benefit. Retraining takes time (1–3 years for technical roles) — meanwhile workers face unemployment and falling incomes. Geographic mismatch — new green jobs may be in different regions from old fossil fuel jobs. Older workers face barriers to reskilling and may be reluctant to switch.

Energy rebates limitations: Don't address the underlying structural problem of inequality — workers remain unemployed or in low-pay jobs. Permanent fiscal burden as long as price gap persists. May reduce incentive to invest in energy efficiency. Can be regressive if poorly designed (universal rebates benefit higher earners more).

Judgement: For long-term inequality reduction, retraining is more effective because it tackles the root cause — workers' position in the labour market. Energy rebates are better for immediate crisis support and political acceptability. The best 'just transition' policy combines both: short-term rebates to protect vulnerable households during transition, plus retraining to enable workers to access the new green economy. Neither alone is sufficient.